Single-Image SAR Training Sample Sweep

Objective

This experiment tests how AlphaEarth-to-Sentinel-1 SAR prediction quality changes as more valid image pixels are used for supervised training.

Unlike the reconstruction report, this run does not write a reconstructed SAR image. It only trains models and evaluates R^2, RMSE, and MAE on one fixed validation set.

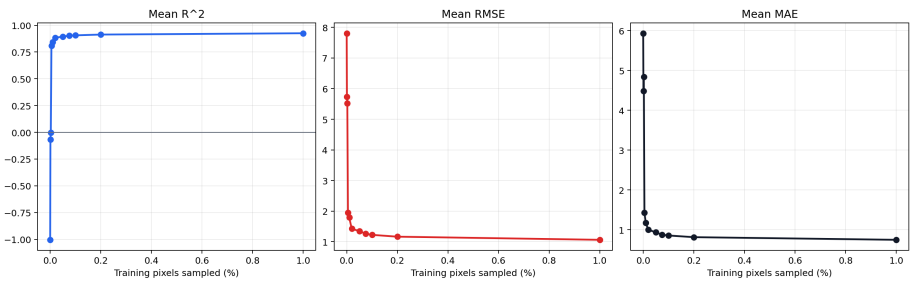
Setup

- Input stack: `/Users/cassius/Github-workspace/alphaearth-to-sar-cross-model-prediction/Data`
- Valid pixels: `1265192`
- Validation fraction: `0.200`
- Validation rows: `253038`
- Candidate training pool rows: `1012154`
- Predictors: `A00` to `A63`
- Targets: `S1_VV`, `S1_VH`, `S1_VV_div_VH`
- Model for nonzero sample rates: `MultiOutputRegressor(LGBMRegressor)`

The `0.0%` row is a no-training random baseline. It predicts each validation pixel by drawing random SAR values from the non-validation pool, so it measures chance-level performance rather than a learned AlphaEarth-to-SAR mapping.

Mean Metrics Across SAR Bands

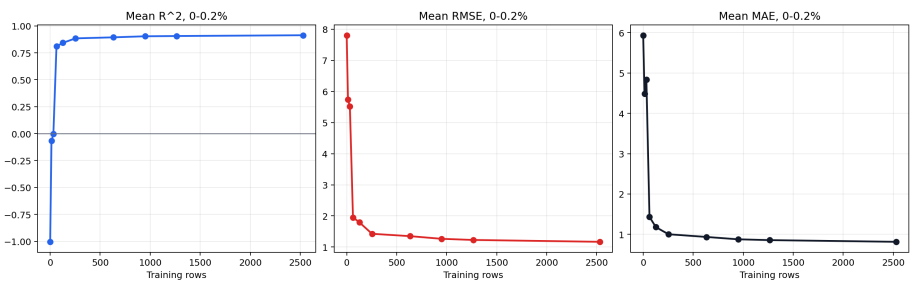
Sample %	Train rows	Mean R^2	Mean RMSE	Mean MAE
0	0	-1.005	7.8075	5.9356
0.001	13	-0.068	5.7368	4.4883
0.0025	32	-0.002	5.5157	4.8369
0.005	63	0.810	1.9505	1.4321
0.01	127	0.843	1.7930	1.1783
0.02	253	0.884	1.4253	1.0019
0.05	633	0.894	1.3489	0.9347
0.075	949	0.904	1.2635	0.8765
0.1	1265	0.906	1.2267	0.8561
0.2	2530	0.913	1.1670	0.8143
1.0	12652	0.926	1.0646	0.7478



Low-Rate Detail

The largest mean R^2 jump in this run occurs between `0%` (0 rows) and `0.001%` (13 rows): mean R^2 increases by `0.937`.

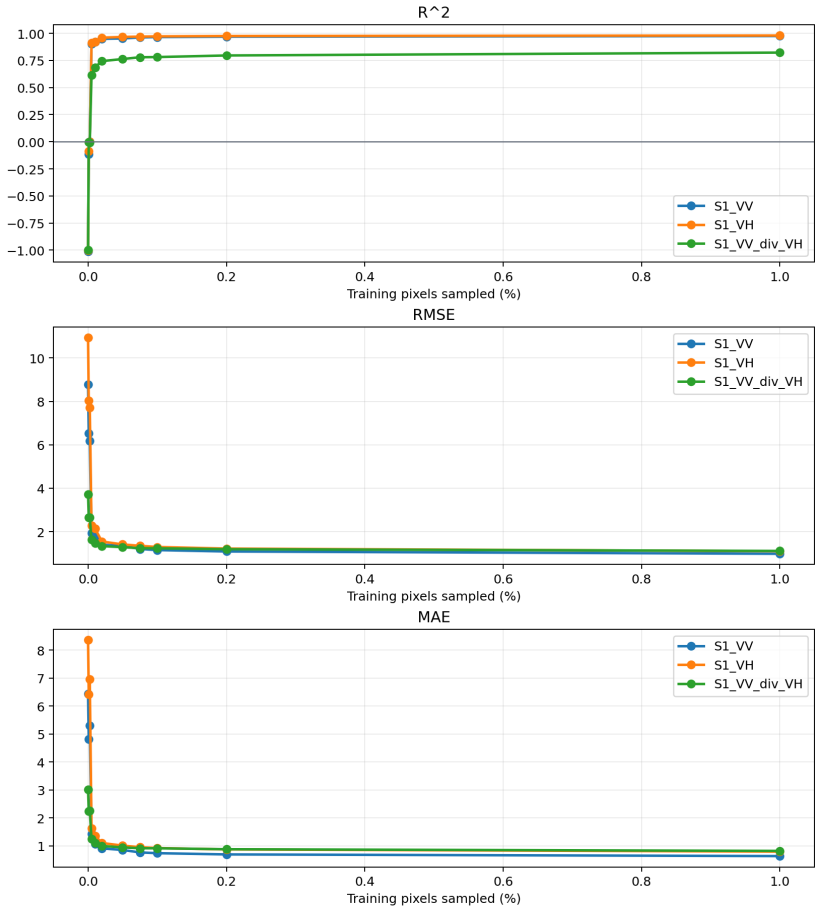
Among learned models only, the sharpest increase occurs between `0.0025%` (32 rows) and `0.005%` (63 rows): mean R^2 increases by `0.812`.



Metrics By SAR Band

Sample %	Train rows	Band	R^2	RMSE	MAE
0	0	`S1_VV`	-1.012	8.7681	6.4371
0	0	`S1_VH`	-1.004	10.9248	8.3526
0	0	`S1_VV_div_VH`	-0.998	3.7297	3.0171
0.001	13	`S1_VV`	-0.115	6.5257	4.8038
0.001	13	`S1_VH`	-0.086	8.0410	6.4175
0.001	13	`S1_VV_div_VH`	-0.004	2.6437	2.2434
0.0025	32	`S1_VV`	-0.000	6.1811	5.3009
0.0025	32	`S1_VH`	-0.001	7.7190	6.9483
0.0025	32	`S1_VV_div_VH`	-0.007	2.6471	2.2616
0.005	63	`S1_VV`	0.902	1.9344	1.4201
0.005	63	`S1_VH`	0.913	2.2808	1.6264
0.005	63	`S1_VV_div_VH`	0.615	1.6363	1.2499
0.01	127	`S1_VV`	0.920	1.7479	1.0671
0.01	127	`S1_VH`	0.922	2.1521	1.3584
0.01	127	`S1_VV_div_VH`	0.686	1.4788	1.1094

Sample %	Train rows	Band	R^2	RMSE	MAE
0.02	253	`S1_VV`	0.949	1.3944	0.9087
0.02	253	`S1_VH`	0.960	1.5454	1.1017
0.02	253	`S1_VV_div_VH`	0.744	1.3361	0.9953
0.05	633	`S1_VV`	0.952	1.3494	0.8518
0.05	633	`S1_VH`	0.966	1.4124	1.0079
0.05	633	`S1_VV_div_VH`	0.763	1.2850	0.9442
0.075	949	`S1_VV`	0.962	1.2031	0.7643
0.075	949	`S1_VH`	0.969	1.3482	0.9523
0.075	949	`S1_VV_div_VH`	0.779	1.2393	0.9129
0.1	1265	`S1_VV`	0.965	1.1510	0.7370
0.1	1265	`S1_VH`	0.972	1.2941	0.9196
0.1	1265	`S1_VV_div_VH`	0.781	1.2350	0.9117
0.2	2530	`S1_VV`	0.969	1.0876	0.6926
0.2	2530	`S1_VH`	0.975	1.2208	0.8717
0.2	2530	`S1_VV_div_VH`	0.796	1.1926	0.8786
1.0	12652	`S1_VV`	0.975	0.9815	0.6331
1.0	12652	`S1_VH`	0.980	1.0998	0.7920
1.0	12652	`S1_VV_div_VH`	0.822	1.1126	0.8182



Interpretation

R^2 should increase and RMSE/MAE should decrease as training pixels are added if the embeddings contain predictive SAR information. The random baseline is expected to have strongly negative R^2 because it draws unrelated SAR values from the scene distribution instead of using AlphaEarth features.